

Ben Hawk

CS 4700 Introduction to Database Management Systems

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Hospital Database

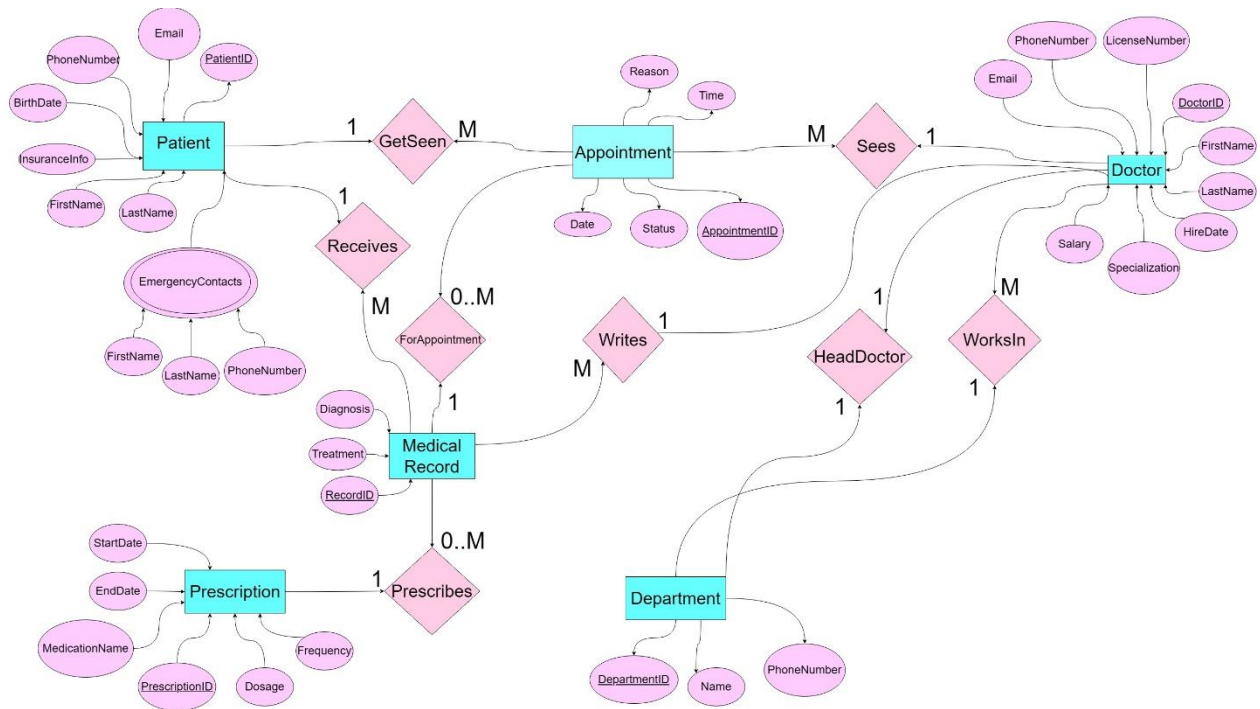
December 5th, 2025

Overview:

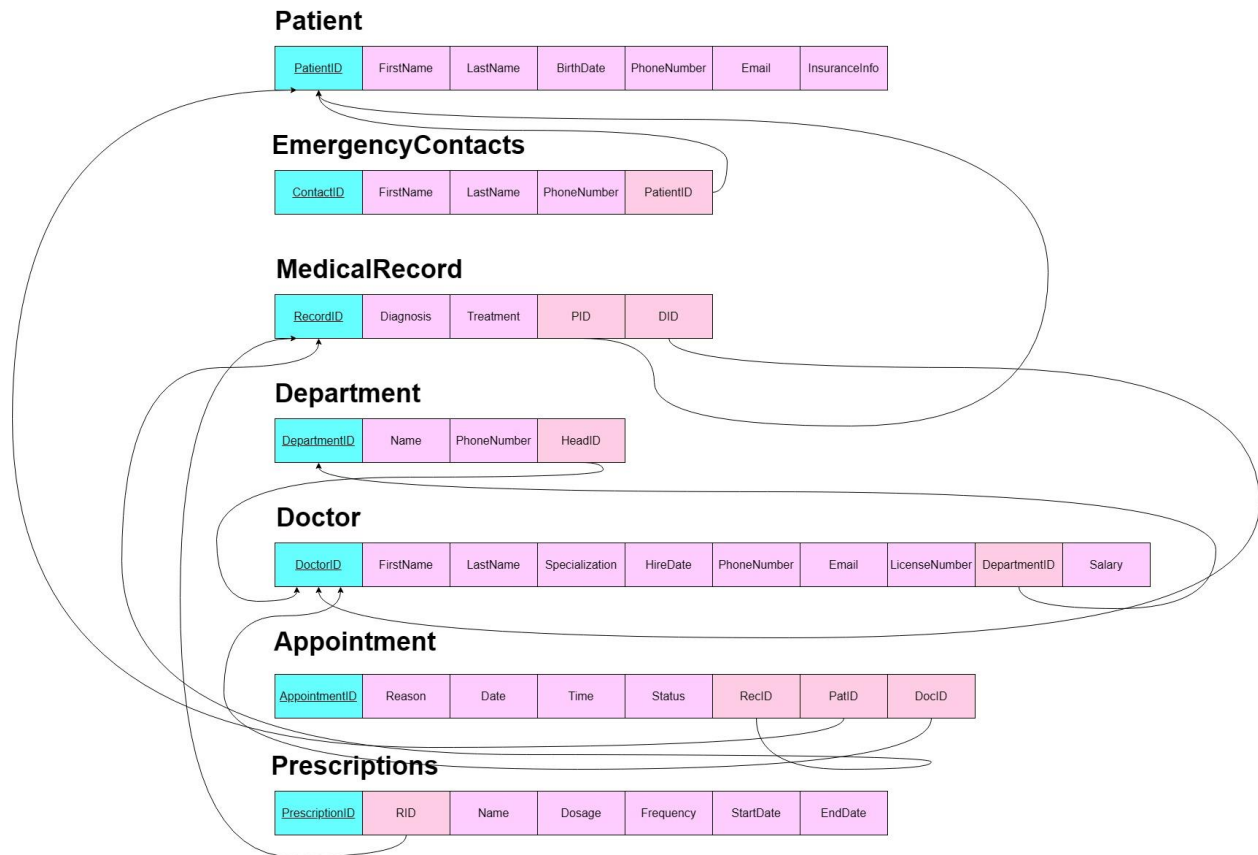
This database is designed to be used in a hospital setting and is primarily focused on the interaction between the given Hospital's Departments, their Doctors, and the Patient's Medical Records and Appointments. The entity types are as follows:

- Patient – this entity type contains a Patient's first and last name, as well as their birth date, their phone number and email address, their insurance status, as well as any number of emergency contacts they might have. The emergency contacts include the names of each contact and their phone number.
- Department – this entity type details the various departments the hospital may have, including their name, ID, and the number to contact them.
- Doctor – this entity type includes the doctor's first and last name, the field they are most proficient in, the date they were hired, the work email address and phone number they were assigned, as well as their license number and salary. Every department can have many doctors who work for it, but there is only one head doctor assigned to each department. Each doctor's salary cannot be lower than \$50,000, and the salary of the head doctor of a given department should be greater than or equal to any other doctors in that department.
- Medical Record – The medical record entity type includes the record ID, the diagnosis of a patient, the treatment decided for the diagnosis, as well as the possible prescriptions chosen. Each patient can have many medical records and each doctor can write many records.
- Appointment – the appointment entity type includes the reason for the appointment, the scheduled time and date, as well as whether the appointment has been cancelled, scheduled, completed, or if the patient did not show up. Each appointment is associated with one medical record, but a medical record can exist for zero appointments or many.
- Prescription – the prescription entity type includes the medication name, the dosage, the frequency, the start date, and the end date. The start date cannot be greater than or the same as the end date. Each medical record can have many prescriptions or none at all.

ER Diagram



Relational Diagram:



Data Requirements

- Patient
 - PatientID – primary key, auto incrementing int, cannot be null
 - FirstName – varchar(255)
 - LastName – varchar(255)
 - BirthDate – date
 - PhoneNumber – varchar(15), has unique constraint
 - Email – varchar(30), has unique constraint
 - InsuranceInfo – enum('Not Insured', 'Insured') with the default being 'Not Insured'.
- Department
 - DepartmentID – primary key, auto incrementing int, cannot be null
 - Name – varchar(255)
 - PhoneNumber – varchar(255)

- HeadID – int, foreign key referencing DoctorID from Doctor table, set null on delete. This can be null in transition periods between head doctors. Set to cascade on update. Has unique constraint.
- Doctor
 - DoctorID – primary key, auto incrementing int, cannot be null
 - FirstName – varchar(255)
 - LastName – varchar(255)
 - Specialization – varchar(255)
 - HireDate – date
 - PhoneNumber – varchar(15), has unique constraint
 - Email – varchar(255), has unique constraint
 - LicenseNumber – int, has unique constraint
 - DepartmentID – int, foreign key referencing DepartmentID from Department table, set null on delete for when departments are changed. Set to cascade on update.
 - Salary – int with default value of 50,000. Has check constraint to be greater than or equal to 50,000, as well as insert and update triggers to ensure that no doctor who is not the head doctor of a given department has a greater salary than the head doctor.
- Medical Record
 - RecordID – primary key, auto incrementing int, cannot be null
 - Diagnosis – varchar(255)
 - Treatment – varchar(255)
 - PID – int, foreign key referencing PatientID from Patient table, set to cascade on delete for when Patients are no longer in the hospital's database, set to cascade on update.
 - DID – int, foreign key referencing DoctorID from Doctor table, set to null on delete in the case that a Doctor no longer works at the hospital, as the patient would still need a medical record. Set to cascade on update.
- Appointment
 - AppointmentID – primary key, auto incrementing int, cannot be null
 - Reason – varchar(255)
 - Date – date
 - Time – time
 - Status – enum('Scheduled', 'Completed', 'Cancelled', 'No-Show'), will default to 'Scheduled'
 - RecID – int, foreign key referencing RecordID from the Medical Record table, set to cascade on delete in the case that a medical record is deleted, and set to cascade on update.

- PatID – int, foreign key referencing PatientID from the Patient table, set to cascade on delete and cascade on update
- DocID – int, foreign key referencing DoctorID from the Doctor table, set to set null on delete, and cascade on update
- Has a unique constraint index on Date, Time, and PatID to prevent patients from having overlapping appointments
- Has a unique constraint index on Date, Time, and DocID to prevent doctors from being scheduled for more than one patient at a given time
- Emergency Contacts
 - ContactID – primary key, auto incrementing int, cannot be null
 - FirstName – varchar(255)
 - LastName – varchar(255)
 - PhoneNumber – varchar(15), has a unique constraint
 - PatientID – int, foreign key referencing PatientID from the Patient table, set to cascade on delete in case a patient is no longer in the database, and is set to cascade on update.
- Prescription
 - PrescriptionID – primary key, auto incrementing int, cannot be null
 - RecordID – int, foreign key referencing RecordID from the MedicalRecord table, set to cascade on delete, set to cascade on update
 - MedicationName – varchar(100)
 - Dosage – varchar(50)
 - Frequency – varchar(100)
 - StartDate – date
 - EndDate – date
 - To prevent patients from being prescribed the same medication more than once in a given time frame, there is a unique constraint index on RecordID, MedicationName, Dosage, Frequency, and StartDate
 - There is a constraint to check that StartDate is less than EndDate, as well as a constraint to check that the two attributes are not equal.

Implementation with MySQL

For this database, the following tables and tuples are being used:

- Patient

PatientID	FirstName	LastName	BirthDate	PhoneNumber	Email	InsuranceInfo
1	Johnny	Silverhand	2003-05-19	447-278-8888	silverhand@gmail.com	Insured
2	Mr	Beast	1998-08-25	442-191-1999	beast@yahoo.com	Not Insured
3	Oh-Long	Johnson	1956-04-23	459-234-3459	johnson@wright.edu	Insured
4	Vro	Meimei	1956-04-24	459-239-3888	vromeimei@yahoo.com	Not Insured
5	John	Dark-Souls	2011-10-04	378-229-3548	miyazaki@gmail.com	Insured
6	John	Halo	2011-10-04	848-929-1283	chief@gmail.com	Insured
7	Syber	Poonk	2008-10-04	838-536-9918	punk77@yahoo.com	Not Insured
8	Bruce	Wayne	1988-11-04	353-292-1818	wayne@wayne.com	Not Insured
9	Obama	Obama	2008-11-05	317-272-8888	barack@whitehouse.com	Insured
10	Mikael	Akerfeldt	1999-11-08	888-939-888	sorrow@gmail.com	Insured
11	Last	Christmas	2000-12-25	999-999-9999	gaveyoumyheart@gmail.com	Insured

- Doctor

DoctorID	FirstName	LastName	Specialization	HireDate	PhoneNumber	Email	LicenseNumber	DepartmentID	Salary
1	Peta	Attrics	Pediatrics	2012-05-23	937-327-1000	peta@ohclinic.com	1023498981	1	200000
2	Aya	Moore-Jenncy	Emergency	2013-08-11	937-327-1001	aya@ohclinic.com	1435658888	6	300000
3	Rad	Aulogee	Radiology	2013-08-11	937-327-1002	rad@ohclinic.com	1534898945	3	500000
4	Barack	Obama	Emergency	2013-09-11	937-327-1003	barack@ohclinic.com	1893456889	6	400000
5	Aycee	Youh	Intensive Care	2014-02-25	937-327-1004	aycee@ohclinic.com	1324789872	2	300000
6	Arnold	Peetee	Physical Therapy	2014-03-28	937-327-1005	arnold@ohclinic.com	1976355687	4	120000
7	Mehd	Serge-Rey	Medical Surgery	2014-04-23	937-327-1006	mehd@ohclinic.com	1342278657	5	450000
8	Messus	Beezt	OB/GYN	2015-03-22	937-327-1007	messus@ohclinic.com	1576667891	7	375000
9	Aych	Bahmb	Radiology	2008-09-05	937-327-9995	aych@ohclinic.com	1883872345	3	400000
10	Boore-rock	Ahbohma	OB/GYN	2008-10-06	937-327-9996	boore@ohclinic.com	1889991113	7	275000
11	John	Physical Therapy	Physical Therapy	2009-11-07	937-327-9999	john@ohclinic.com	1929456977	4	100000

- Department

DepartmentID	Name	PhoneNumber	HeadID
1	Pediatrics	937-227-1000	1
2	Intensive Care Unit	937-227-1001	5
3	Radiology	937-227-1002	3
4	Physical Therapy	937-227-1003	6
5	Medical-Surgical Unit	937-227-1004	7
6	Emergency Department	937-227-1005	4
7	OB/GYN	937-227-1006	8

- MedicalRecord

RecordID	Diagnosis	Treatment	PID	DID
1	Rhinovirus	Medication and Rest	5	1
2	Minor Scoliosis	BiWeekly PT Sessions and pain relief medication as needed	4	6
3	Grade 1 Glioma	Surgery	1	7
4	Epilepsy	Stereotactic Radiosurgery	3	3
5	Hip Impingement	PT Sessions	2	6
6	Covid-19	Antiviral medication	6	1

- Appointment

AppointmentID	Reason	Date	Time	Status	RecID	PatID	DocID
1	Physical Therapy	2025-11-08	13:30:00	Scheduled	2	4	6
2	Physical Therapy	2025-11-07	14:00:00	Completed	5	2	6
3	Epilepsy Treatment	2025-12-03	15:25:00	Scheduled	4	3	3
4	Common Cold	2025-05-04	10:30:00	No-Show	1	5	1
5	Tumor Removal	2025-12-05	11:45:00	Scheduled	3	1	7
6	Physical Therapy	2025-01-05	11:00:00	Cancelled	5	2	6
7	Physical Therapy	2025-03-06	11:30:00	Cancelled	5	2	6
8	Physical Therapy	2025-05-10	12:00:00	No-Show	5	2	6
9	Coronavirus	2020-10-01	12:00:00	Completed	6	6	1

- EmergencyContacts

ContactID	FirstName	LastName	PhoneNumber	PatientID
1	Geralt	Of-Rivia	787-993-8949	1
2	V	Punk	993-484-9870	1
3	CD	Projekt	839-789-1000	1
4	Tole	Tole	839-890-1001	4
5	Opeth	Band	764-901-3001	10
6	Bat	Man	111-111-1111	8
7	Alfred	Pennyworth	112-111-1111	8
8	Santa	Claus	333-333-3334	11

- Prescription

PrescriptionID	RID	MedicationName	Dosage	Frequency	StartDate	EndDate
1	4	Brivaracetam	50 mg	Twice a day	2025-11-07	2026-11-07
2	6	Nirmatrelvir	150 mg	Twice a day	2020-10-01	2020-10-06
3	6	Ritonavir	100 mg	Twice a day	2020-10-01	2020-10-06
4	3	Morphine	2.5 mg	Every four hours	2025-12-05	2020-12-12

Queries

SELECT

- WHERE:
 - A simple query to retrieve all Patients with even PatientIDs

```
mysql> SELECT * FROM Patient
-> WHERE PatientID % 2 = 0;
```

PatientID	FirstName	LastName	BirthDate	PhoneNumber	Email	InsuranceInfo
2	Mr	Beast	1998-08-25	442-191-1999	beast@yahoo.com	Not Insured
4	Vro	Meimei	1956-04-24	459-239-3888	vromeimei@yahoo.com	Not Insured
6	John	Halo	2011-10-04	848-929-1283	chief@gmail.com	Insured
8	Bruce	Wayne	1988-11-04	353-292-1818	wayne@wayne.com	Not Insured
10	Mikael	Akerfeldt	1999-11-08	888-939-888	sorrow@gmail.com	Insured

```
5 rows in set (0.01 sec)
```

- Unspecified WHERE
 - A query to retrieve all emails from the Patient table

```
mysql> SELECT Email
-> FROM Patient;
```

Email
barack@whitehouse.com
beast@yahoo.com
chief@gmail.com
gaveyoumyheart@gmail.com
johnson@wright.edu
miyazaki@gmail.com
punk77@yahoo.com
silverhand@gmail.com
sorrow@gmail.com
vromeimei@yahoo.com
wayne@wayne.com

```
11 rows in set (0.00 sec)
```

SELECT-JOIN-PROJECT

- Without Aliases:
 - This query finds the insurance info of each patient, as well as any medicine they have been prescribed and its associated medical record ID

```
mysql> SELECT MedicationName, InsuranceInfo, PatientID, RecordID
-> FROM Patient, Prescription, MedicalRecord
-> WHERE PatientID=PID AND RecordID=RID;
```

MedicationName	InsuranceInfo	PatientID	RecordID
Morphine	Insured	1	3
Brivaracetam	Insured	3	4
Nirmatrelvir	Insured	6	6
Ritonavir	Insured	6	6

4 rows in set (0.00 sec)

- With Aliases:
 - This query retrieves the first and last name of all head doctors who have an appointment currently scheduled.

```
mysql> SELECT FirstName, LastName
-> FROM Doctor D, Department O, Appointment A
-> WHERE D.DoctorID=O.HeadID AND D.DoctorID=A.DocID AND A.Status='Scheduled';
```

FirstName	LastName
Arnold	Peetee
Rad	Aulogee
Mehd	Serge-Rey

3 rows in set (0.00 sec)

RENAMING

- This query retrieves the first and last name of each head doctor who has an appointment scheduled renamed to DoctorFname and DoctorLname, as well as the department name renamed to DepartmentName and the reason for the appointment renamed to AppointmentReason.

```
mysql> SELECT D.FirstName AS DoctorFname, D.LastName AS DoctorLname, O.Name AS DepartmentName, A.Reason AS AppointmentReason
-> FROM Doctor AS D, Department as O, Appointment AS A
-> WHERE D.DoctorID=O.HeadID AND D.DoctorID=A.DocID AND A.Status='Scheduled';
```

DoctorFname	DoctorLname	DepartmentName	AppointmentReason
Arnold	Peetee	Physical Therapy	Physical Therapy
Rad	Aulogee	Radiology	Epilepsy Treatment
Mehd	Serge-Rey	Medical-Surgical Unit	Tumor Removal

3 rows in set (0.00 sec)

UNION

- This query retrieves the PatientID from patients who have emergency contacts as well or who are not insured.

```
mysql> (SELECT P.PatientID FROM Patient AS P, Emergencycontacts AS E WHERE P.PatientID=E.PatientID)
-> UNION
-> (SELECT P.PatientID FROM Patient AS P WHERE InsuranceInfo='Not Insured');
+-----+
| PatientID |
+-----+
| 1 |
| 4 |
| 8 |
| 10 |
| 11 |
| 2 |
| 7 |
+-----+
7 rows in set (0.00 sec)
```

SUBSTRING

- This query retrieves the first name, last name, and email address of all patients with an email using the gmail.com domain name.

```
mysql> SELECT FirstName, LastName, Email
-> FROM Patient
-> WHERE Email LIKE '%@gmail.com%';
+-----+-----+-----+
| FirstName | LastName | Email |
+-----+-----+-----+
| Johnny | Silverhand | silverhand@gmail.com |
| John | Dark-Souls | miyazaki@gmail.com |
| John | Halo | chief@gmail.com |
| Mikael | Akerfeldt | sorrow@gmail.com |
| Last | Christmas | gaveyoumyheart@gmail.com |
+-----+-----+-----+
5 rows in set (0.01 sec)
```

ARITHMETIC

- Addition
 - This query increases all head doctor's salary by \$10,000 and retrieves their first and last name as well as their increased salary.

```
mysql> SELECT FirstName, LastName, Salary+10000 AS IncreasedSalary
-> FROM Doctor, Department
-> WHERE DoctorID=HeadID;
```

FirstName	LastName	IncreasedSalary
Peta	Attrics	210000
Rad	Aulogee	510000
Barack	Obama	410000
Aycee	Youh	310000
Arnold	Peetee	130000
Mehd	Serge-Rey	460000
Messus	Beezt	385000

7 rows in set (0.01 sec)

- Subtraction
 - This query reduces the salary of doctors who currently have a scheduled appointment by \$20000 and retrieves their first name, last name, and salary renamed to ReducedSalary.

```
mysql> SELECT FirstName, LastName, Salary-20000 AS ReducedSalary
-> FROM Doctor, Appointment
-> WHERE DoctorID = DocID AND Status='Scheduled';
```

FirstName	LastName	ReducedSalary
Arnold	Peetee	100000
Rad	Aulogee	480000
Mehd	Serge-Rey	430000

3 rows in set (0.00 sec)

- Multiplication
 - This query retrieves the first and last name as well as a doubled salary named SalaryTimesTwo from the doctors who are not head of a department.

```
mysql> SELECT FirstName, LastName, Salary*2 AS SalaryTimesTwo
-> FROM
-> ((SELECT FirstName, LastName, Salary FROM Doctor)
-> EXCEPT
-> (SELECT FirstName, LastName, Salary FROM Doctor, Department WHERE DoctorID=HeadID)) AS NotHeadDoctors;
```

FirstName	LastName	SalaryTimesTwo
Aya	Moore-Jenncy	600000
Aych	Bahmb	800000
Boore-rock	Ahbohama	550000
John	Physical Therapy	200000

4 rows in set (0.00 sec)

- Division
 - This query retrieves the first and last name from all doctors as well as their salary divided by two, renamed to DividedSalary.

```
mysql> SELECT FirstName, LastName, Salary/2 AS DividedSalary
-> FROM Doctor;
```

FirstName	LastName	DividedSalary
Peta	Attrics	100000.0000
Aya	Moore-Jenncy	150000.0000
Rad	Aulogee	250000.0000
Barack	Obama	200000.0000
Aycee	Youh	150000.0000
Arnold	Peetee	60000.0000
Mehd	Serge-Rey	225000.0000
Messus	Beezt	187500.0000
Aych	Bahmb	200000.0000
Boore-rock	Ahbohama	137500.0000
John	Physical Therapy	50000.0000

11 rows in set (0.00 sec)

ORDER BY

- This query retrieves the first and last name of all patients with a medical record and sorts them by their last name.

```
mysql> SELECT P.FirstName, P.LastName
-> FROM Patient P, MedicalRecord R
-> WHERE P.PatientID = R.PID
-> ORDER BY P.LastName, P.FirstName;

+-----+-----+
| FirstName | LastName |
+-----+-----+
| Mr       | Beast   |
| John     | Dark-Souls |
| John     | Halo    |
| Oh-Long  | Johnson |
| Vro      | Meimei  |
| Johnny   | Silverhand |
+-----+-----+
6 rows in set (0.00 sec)
```

DELETE

- The query deletes the doctor with a DoctorID of 8, making the HeadID of the OB/GYN temporarily null due to its on delete rule.

```
mysql> DELETE FROM Doctor
-> WHERE DoctorID = 8;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Department;

+-----+-----+-----+-----+
| DepartmentID | Name                | PhoneNumber | HeadID |
+-----+-----+-----+-----+
| 1 | Pediatrics          | 937-227-1000 | 1 |
| 2 | Intensive Care Unit | 937-227-1001 | 5 |
| 3 | Radiology           | 937-227-1002 | 3 |
| 4 | Physical Therapy    | 937-227-1003 | 6 |
| 5 | Medical-Surgical Unit | 937-227-1004 | 7 |
| 6 | Emergency Department | 937-227-1005 | 4 |
| 7 | OB/GYN              | 937-227-1006 | NULL |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM Doctor;

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| DoctorID | FirstName | LastName | Specialization | HireDate | PhoneNumber | Email | LicenseNumber | DepartmentID | Salary |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Peta | Attrics | Pediatrics | 2012-05-23 | 937-327-1000 | peta@ohclinic.com | 1023498981 | 1 | 200000 |
| 2 | Aya | Moore-Jenncy | Emergency | 2013-08-11 | 937-327-1001 | aya@ohclinic.com | 1435658888 | 6 | 300000 |
| 3 | Rad | Aulogee | Radiology | 2013-08-11 | 937-327-1002 | rad@ohclinic.com | 1534898945 | 3 | 500000 |
| 4 | Barack | Obama | Emergency | 2013-09-11 | 937-327-1003 | barack@ohclinic.com | 1893456889 | 6 | 400000 |
| 5 | Aycee | Youh | Intensive Care | 2014-02-25 | 937-327-1004 | aycee@ohclinic.com | 1324789872 | 2 | 300000 |
| 6 | Arnold | Peetee | Physical Therapy | 2014-03-28 | 937-327-1005 | arnold@ohclinic.com | 1976355687 | 4 | 120000 |
| 7 | Mehd | Serge-Rey | Medical Surgery | 2014-04-23 | 937-327-1006 | mehd@ohclinic.com | 1342278657 | 5 | 450000 |
| 9 | Aych | Bahmb | Radiology | 2008-09-05 | 937-327-9995 | aych@ohclinic.com | 1883872345 | 3 | 400000 |
| 10 | Boore-rock | Ahbohma | OB/GYN | 2008-10-06 | 937-327-9996 | boore@ohclinic.com | 1889991113 | 7 | 275000 |
| 11 | John | Physical Therapy | Physical Therapy | 2009-11-07 | 937-327-9999 | john@ohclinic.com | 1929456977 | 4 | 100000 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

UPDATE

- These queries updates the OB/GYN HeadID to 10, and the salary of the doctor to 350000 to match their position.

```
mysql> UPDATE Department
-> SET HeadID = 10
-> WHERE DepartmentID = 7
-> ;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Doctor
-> SET Salary=350000
-> WHERE DoctorID = 10;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Department
-> WHERE DepartmentID = 7;
+-----+-----+-----+-----+
| DepartmentID | Name   | PhoneNumber | HeadID |
+-----+-----+-----+-----+
| 7           | OB/GYN | 937-227-1006 | 10     |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM Doctor WHERE DoctorID = 10;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| DoctorID | FirstName | LastName | Specialization | HireDate | PhoneNumber | Email | LicenseNumber | DepartmentID | Salary |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 10       | Boore-rock | Ahbohama | OB/GYN         | 2008-10-06 | 937-327-9996 | boore@ohclinic.com | 1889991113 | 7           | 350000 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

COMPARISON OPERATORS

- IN
 - This query retrieves the first name, last name, and insurance info of patients who were born in November or have a no-show appointment.

```
mysql> SELECT DISTINCT FirstName, LastName, InsuranceInfo
-> FROM Patient
-> WHERE PatientID IN
-> ((SELECT PatientID FROM Patient WHERE BirthDate LIKE '%-11-%')
-> UNION
-> (SELECT PatientID FROM Patient, Appointment WHERE PatientID=PatID AND Status='No-Show'));
+-----+-----+-----+
| FirstName | LastName | InsuranceInfo |
+-----+-----+-----+
| Mr        | Beast    | Not Insured   |
| John      | Dark-Souls | Insured       |
| Bruce     | Wayne    | Not Insured   |
| Ohbama    | Obama    | Insured       |
| Mikael    | Akerfeldt | Insured       |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

- NOT IN
 - This query retrieves the first name, last name, and salary of all doctors who do not have any current appointments or do not have a salary greater than 300000.

```
mysql> SELECT DISTINCT FirstName, LastName, Salary
-> FROM Doctor
-> WHERE DoctorID NOT IN
-> ((SELECT DoctorID FROM Doctor WHERE Salary > 300000)
-> UNION
-> (SELECT DoctorID FROM Doctor, Appointment WHERE DoctorID = DocID));
```

FirstName	LastName	Salary
Aya	Moore-Jenncy	300000
Aycee	Youh	300000
John	Physical Therapy	100000

3 rows in set (0.01 sec)

- =
 - This query retrieves all doctors making exactly \$300,000 salary.

```
mysql> SELECT * FROM Doctor
-> WHERE Salary = 300000;
```

DoctorID	FirstName	LastName	Specialization	HireDate	PhoneNumber	Email	LicenseNumber	DepartmentID	Salary
2	Aya	Moore-Jenncy	Emergency	2013-08-11	937-327-1001	aya@ohclinic.com	1435658888	6	300000
5	Aycee	Youh	Intensive Care	2014-02-25	937-327-1004	aycee@ohclinic.com	1324789872	2	300000

2 rows in set (0.00 sec)

- >=
 - This query retrieves all doctors making at least \$500,000 salary.

```
mysql> SELECT FirstName, LastName, Specialization, Salary
-> FROM Doctor
-> WHERE Salary >= 500000;
```

FirstName	LastName	Specialization	Salary
Rad	Aulogee	Radiology	500000

1 row in set (0.00 sec)

- <
 - This query retrieves the first name, last name, specialization, salary of doctors making less than doctors who specialize in Physical Therapy.

```
mysql> SELECT FirstName, LastName, Specialization, Salary
-> FROM Doctor
-> WHERE Salary < ALL (SELECT Salary FROM Doctor WHERE Specialization = 'Physical Therapy');
```

Empty set (0.01 sec)

- >

- This query retrieves the first name, last name, specialization, and salary from doctors making more than all doctors hired in 2008.

```
mysql> SELECT FirstName, LastName, Specialization, Salary
-> FROM Doctor
-> WHERE Salary > ALL (SELECT Salary FROM Doctor WHERE HireDate LIKE '%2008%');
```

FirstName	LastName	Specialization	Salary
Rad	Aulogee	Radiology	500000
Mehd	Serge-Rey	Medical Surgery	450000

2 rows in set (0.00 sec)

- <=

- This query retrieves the first name and ID of doctors making less than or equal to \$250,000

```
mysql> SELECT DoctorID, FirstName
-> FROM Doctor
-> WHERE Salary <= 250000;
```

DoctorID	FirstName
1	Peta
6	Arnold
11	John

3 rows in set (0.00 sec)

- <>

- This query retrieves the doctor ID and first name of doctors whose salary is not equal to all doctors specializing in Emergency.

```
mysql> SELECT DoctorID, FirstName
-> FROM Doctor
-> WHERE Salary <> ALL (Select Salary FROM Doctor WHERE Specialization='Emergency');
```

DoctorID	FirstName
1	Peta
3	Rad
6	Arnold
7	Mehd
10	Boore-rock
11	John

6 rows in set (0.01 sec)

- EXISTS

- This query retrieves the patient ID and first name of patients who have an emergency contact with the same first name.

```
mysql> SELECT PatientID, FirstName
-> FROM Patient AS P
-> WHERE EXISTS
-> (SELECT * FROM EmergencyContacts AS E WHERE P.PatientID = E.PatientID AND P.FirstName = E.FirstName);
Empty set (0.01 sec)
```

- NOT EXISTS

- This query retrieves the patient ID and first name of patients who do not have an emergency contact with the same first name.

```
mysql> SELECT PatientID, FirstName
-> FROM Patient AS P
-> WHERE NOT EXISTS
-> (SELECT * FROM EmergencyContacts AS E WHERE P.PatientID = E.PatientID AND P.FirstName = E.FirstName);
+-----+-----+
| PatientID | FirstName |
+-----+-----+
| 1 | Johnny |
| 2 | Mr |
| 3 | Oh-Long |
| 4 | Vro |
| 5 | John |
| 6 | John |
| 7 | Syber |
| 8 | Bruce |
| 9 | Ohbama |
| 10 | Mikael |
| 11 | Last |
+-----+-----+
11 rows in set (0.00 sec)
```

JOIN

- JOIN

- This query retrieves the first name, salary, and specialization of head doctors who have a scheduled appointment.

```
mysql> SELECT FirstName, Salary, Specialization
-> FROM ((Department JOIN Doctor ON HeadID=DoctorID) JOIN Appointment ON DoctorID=DocID)
-> WHERE Status='Scheduled';
+-----+-----+-----+
| FirstName | Salary | Specialization |
+-----+-----+-----+
| Arnold | 120000 | Physical Therapy |
| Rad | 500000 | Radiology |
| Mehd | 450000 | Medical Surgery |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

- LEFT OUTER JOIN

- This query retrieves the diagnosis, treatment, and prescribed medicine or null if no prescriptions are necessary.

```
mysql> SELECT Diagnosis, Treatment, MedicationName
-> FROM MedicalRecord LEFT OUTER JOIN Prescription ON RecordID=RID;
```

Diagnosis	Treatment	MedicationName
Rhinovirus	Medication and Rest	NULL
Minor Scoliosis	BiWeekly PT Sessions and pain relief medication as needed	NULL
Grade 1 Glioma	Surgery	Morphine
Epilepsy	Stereotactic Radiosurgery	Brivaracetam
Hip Impingement	PT Sessions	NULL
Covid-19	Antiviral medication	Nirmatrelvir
Covid-19	Antiviral medication	Ritonavir

7 rows in set (0.00 sec)

- RIGHT OUTER JOIN

- This query retrieves the diagnosis and treatment as well as the name of the medicine, the dosage, and the frequency to take it for diagnoses that need a prescription.

```
mysql> SELECT Diagnosis, Treatment, MedicationName, Dosage, Frequency
-> FROM MedicalRecord RIGHT OUTER JOIN Prescription ON RecordID=RID;
```

Diagnosis	Treatment	MedicationName	Dosage	Frequency
Grade 1 Glioma	Surgery	Morphine	2.5 mg	Every four hours
Epilepsy	Stereotactic Radiosurgery	Brivaracetam	50 mg	Twice a day
Covid-19	Antiviral medication	Nirmatrelvir	150 mg	Twice a day
Covid-19	Antiviral medication	Ritonavir	100 mg	Twice a day

4 rows in set (0.00 sec)

- FULL OUTER JOIN

- This query retrieves all diagnoses, treatments, as well as the names, dosages, and frequencies of any associated prescriptions for the record, or null if none are associated. As MySQL does not support FULL OUTER JOIN directly, UNION must be used for the LEFT OUTER JOIN and RIGHT OUTER JOIN to get the same result.

```
mysql> (SELECT Diagnosis, Treatment, MedicationName, Dosage, Frequency FROM MedicalRecord LEFT OUTER JOIN Prescription ON RecordID=RID)
-> UNION
-> (SELECT Diagnosis, Treatment, MedicationName, Dosage, Frequency FROM MedicalRecord RIGHT OUTER JOIN Prescription ON RecordID=RID);
```

Diagnosis	Treatment	MedicationName	Dosage	Frequency
Rhinovirus	Medication and Rest	NULL	NULL	NULL
Minor Scoliosis	BiWeekly PT Sessions and pain relief medication as needed	NULL	NULL	NULL
Grade 1 Glioma	Surgery	Morphine	2.5 mg	Every four hours
Epilepsy	Stereotactic Radiosurgery	Brivaracetam	50 mg	Twice a day
Hip Impingement	PT Sessions	NULL	NULL	NULL
Covid-19	Antiviral medication	Nirmatrelvir	150 mg	Twice a day
Covid-19	Antiviral medication	Ritonavir	100 mg	Twice a day

7 rows in set (0.01 sec)

- NATURAL JOIN

- This query retrieves the first name of the patient and the doctor who have matching first names.

```
mysql> SELECT P.FirstName AS PatientFirstName, D.FirstName AS DoctorFirstName
-> FROM (SELECT DISTINCT FirstName FROM Patient) AS P
-> NATURAL JOIN (SELECT DISTINCT FirstName FROM Doctor) AS D;
```

PatientFirstName	DoctorFirstName
John	John

1 row in set (0.00 sec)

AGGREGATE FUNCTIONS

- This query retrieves the count of all Doctors who have a salary, as well as the total being made from all doctors, the maximum salary, the minimum salary, and the average salary.

```
mysql> SELECT COUNT(Salary), SUM(Salary), MAX(Salary), MIN(Salary), AVG(Salary)
-> FROM Doctor;
```

COUNT(Salary)	SUM(Salary)	MAX(Salary)	MIN(Salary)	AVG(Salary)
10	3120000	500000	100000	312000.0000

1 row in set (0.01 sec)

GROUP

- This query retrieves the department ID, the number of doctors in that department, as well as the average salary and groups the resulting table by the department ID.

```
mysql> SELECT DepartmentID, COUNT(*), AVG(Salary)
-> FROM Doctor
-> GROUP BY DepartmentID;
```

DepartmentID	COUNT(*)	AVG(Salary)
1	1	200000.0000
2	1	300000.0000
3	2	450000.0000
4	2	110000.0000
5	1	450000.0000
6	2	350000.0000
7	1	350000.0000

7 rows in set (0.01 sec)

HAVING

- This query retrieves the name and count of any departments that have at least two doctors working in them.

```
mysql> SELECT D.Name, COUNT(*)  
-> FROM Department D, Doctor O  
-> WHERE D.DepartmentID = O.DepartmentID  
-> GROUP BY D.Name  
-> HAVING COUNT(*) >= 2;
```

Name	COUNT(*)
Emergency Department	2
Physical Therapy	2
Radiology	2

3 rows in set (0.00 sec)

CONCLUSION

The database design is primarily focused on the patient's relationships with the other entities. It is efficient at retrieving information relevant to doctors and their patient's medical records, appointments, and any medication they might be prescribed. Some research was done to determine the typical departments present in a hospital, as well as the associated prescription medications and dosages for various diagnoses. Determining the entities to use for this database was relatively fast; however, designing the relationships between them and any constraints they might need to logically function was more time consuming. For example, the Appointment table initially only had a relationship with the MedicalRecord table but was later changed to have foreign keys from the Patient and Doctor tables to more easily detect overlapping appointments. Some future work could be done to include more depth for the hospital, such as other employees like nurses and secretaries and their salaries, as well as more detail over Patient's insurance information and their bills for appointments.